EECS 563 Homework 3

Due: Friday, September 25, 2026, 11:59 pm. Online submission on Canvas.

Problem 1 [8 points]: True or False. Briefly justify your answers.

(a) A user requests a Web page that consists of some text and three images. For this page, the client will send one request message and receive four response messages.

Solution: False. The client will send four requests.

(b) With nonpersistent connections between browser and origin server, it is possible for a single TCP segment to carry two distinct HTTP request messages.

Solution: False. In nonpersistent connection, there is a single TCP connection for each HTTP request.

(c) The `Date:` header in the HTTP response message indicates when the object in the response was last modified.

Solution: False. The "Date:" is the time at which the request was created and not when the object was last modified.

(d) HTTP response messages never have an empty message body.

Solution: False. HTTP uses caching to save bandwidth. Therefore, it is possible for the message body to be empty if the client has a copy of the object that has not been modified. Another instance is when the HTTP request message uses the HEAD method, which requests headers (only) from the HTTP server.

Problem 2 [6 points]: HTTP features and functionalities.

(a) Consider an e-commerce site that wants to keep a purchase record for each of its customers. Describe how this can be done.

Solution: By using Cookies, an HTTP server can keep track of the HTTP clients. When the user first visits the site, the server creates a unique identification number, and it creates an entry in its back-end database, and returns this identification as a cookie to the client. This cookie is stored on the user's host and is managed by the browser. During each subsequent visit (and purchase), the browser sends the cookie back to the site. Thus the site knows when this user (more precisely, this browser) is visiting the site.

(b) Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why?

Solution: Web caching can bring the desired content "closer" to the user, possibly to the same LAN to which the user's host is connected. Web caching can reduce the delay for all objects. If the most up-to-date version of an object is stored in the cache, then the request would not be forwarded to the server. This, in turn, reduces the traffic on the links. Thus, even if an object that is not cached, it will experience a

smaller delay due to reduced traffic on links. Therefore, caching can reduce traffic for all objects requested.

Problem 3 [6 points]: Consider a Web client that wants to retrieve a Web document at a given URL. The IP address of the Web server is initially unknown. List ***all application-layer protocol along with their transport layer protocols*** that are needed in this scenario.

Solution : DNS is used first to resolve the hostname in the URL into the server's IP address, since the IP address is initially unknown. After the IP address is known, the client retrieves the Web document using either HTTP over TCP (for `http://` URLs) or HTTPS over TCP (for `https://` URLs). Please note that we considered HTTP/1 and HTTP/2 in this question.

Application-layer protocol	Transport-layer protocol
DNS	UDP
HTTP	TCP
HTTPS	TCP

Problem 4 [8 points]: Suppose within your Web browser you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your local host, so a DNS lookup is necessary to obtain the IP address. Suppose that N DNS servers are visited before your host receives the IP address from DNS; the successive visits incur the round trip time (RTT) of X_1 , . . .,X_N between your host and the DNS servers. Further, suppose that the Web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Let X_0 denote the RTT between the local host and the web server containing the object.

(a) Describe the steps and protocols involved in receiving the HTML object at each step.

Solution: [4 points] First, DNS will be used to resolve the domain name to IP address. DNS uses UDP as the transport layer protocol. After that, the TCP connection will be established. Finally, the HTTP request is transmitted and the HTTP response is received.

(b) Assuming zero transmission time of the object, calculate the time from when the client clicks on the link until the client receives the object.

Solution: [4 points] The total amount of time to get the IP address is $X_1 + X_2 + ... + X_N$. Once the IP address is known, X_0 elapses to set up the TCP connection and another X_0 elapses to request and receive the small object. The total response time is $2X_0 + X_1 + X_2 + ... + X_N$.

Problem 5 [15 points]: Open Wireshark on your computer and begin a packet capture session. Please refer to homework 2, if you need more details.

In your web browser, visit a website of your choice. You can try websites that you have not visited recently. Stop the Wireshark capture once the page has fully loaded. Filter the captured traffic in Wireshark to display only DNS packets. Use the following filter: `dns` or

`udp.port == 53`. Remember that, by default, DNS uses UDP as the underlying transport layer protocol and port number 53. Identify the DNS query and response packets for the website you visited. Take a screenshot of the DNS query and response packets and include it with your submission. Answer the following questions. Please include screenshots of the Wireshark packet captures.

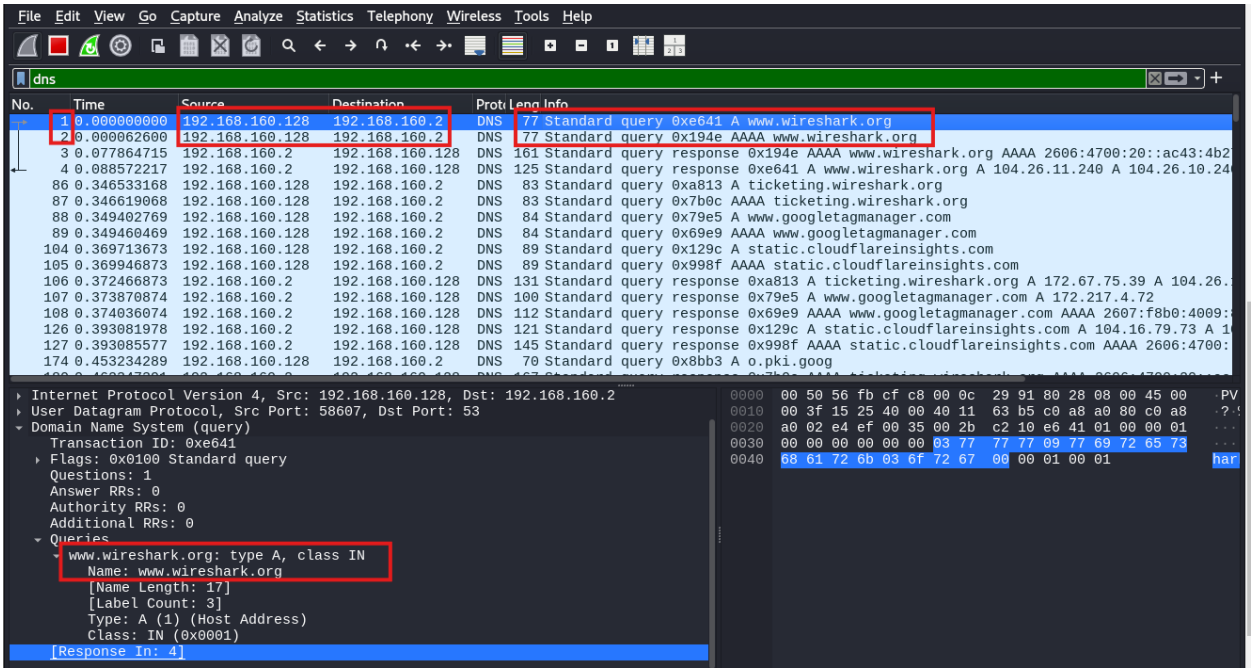


Figure 1: Wireshark Capture for DNS query

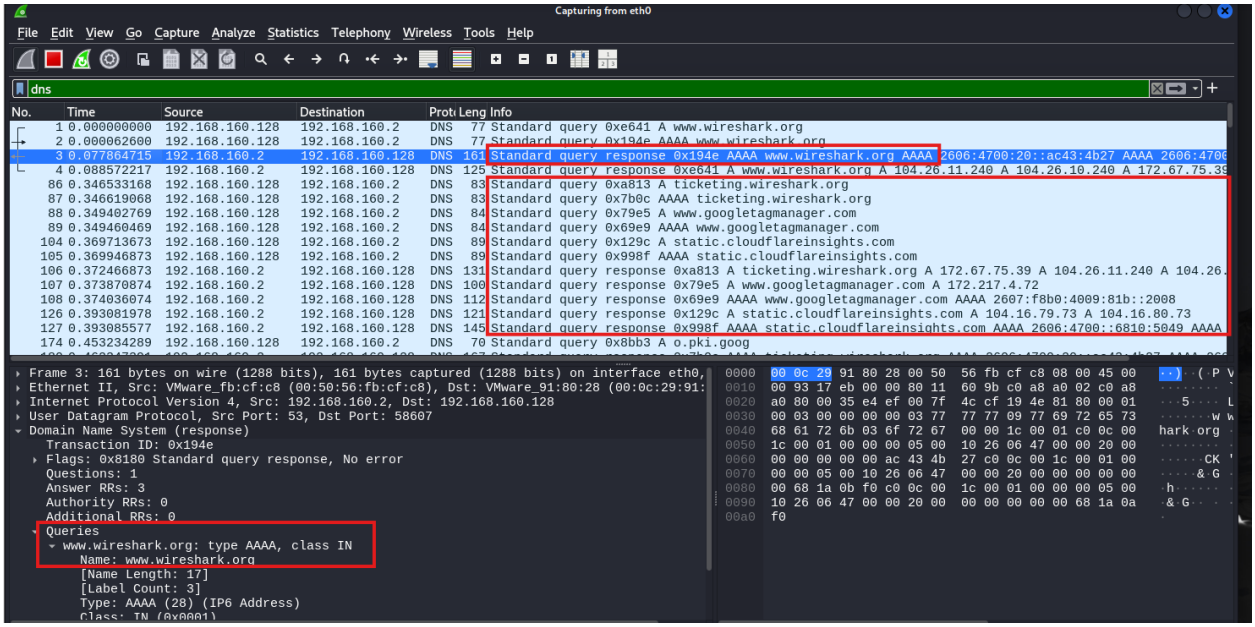


Figure 2: Wireshark Capture for DNS response

(a) What was the domain name in the DNS query?

Solution [3 points]: For this problem, we visited *www.wireshark.org*. As shown in Fig.1, the domain name is *www.wireshark.org*. When you visit a website, there can be multiple DNS queries since the website often loads content from multiple domains and each of these requires a separate DNS query if the domain hasn't been cached.

(b) What is the type of DNS query (e.g., A, AAAA, MX, etc.)? Note that "A" DNS records correspond to IPv4, while "AAAA" type corresponds to IPv6.

Solution [3 points]: Depending on the website you visited your browser may perform DNS queries for different record types (e.g., A, AAAA , etc), which can be verified in Fig.1.

(c) What was the IP address returned in the DNS response?

Solution [3 points]: A DNS response can contain multiple IP addresses for a single domain. As shown in Fig.2, the DNS response returned three IP addresses, 104.26.11.240, 104.26.10.240, 172.67.75.39.

(d) What type of DNS response was provided (e.g., A, AAAA, CNAME, etc.)?

Solution [3 points]: As shown in Fig.2, there could be several DNS responses corresponding to the DNS queries. In this example, we see both type A and type AAAA.

(e) Explain, in your own words, how DNS helps resolve the website's domain name to an IP address based on what you observed in Wireshark.

Solution [3 points]: When you visit a website, your computer doesn't know its IP address. Instead, it relies on the Domain Name System (DNS) to translate the human-readable domain name (e.g., www.wireshark.org) into an IP address (e.g., 104.26.11.240). First, your device (source) (e.g., 192.168.160.128) sends a DNS query packet to a DNS server (192.168.160.2). The query contains "Standard query A www.wireshark.org". Next, the DNS server checks its cache or forwards the query to other DNS servers if necessary. It finds the matching A records and returns one or more IP addresses for the domain. In our example, the response contains "Standard query response A 104.26.11.240 A 104.26.10.240 A 172.67.75.39". Finally, the device picks one of the IPs from the response and initiates a TCP connection to the selected IP to request the actual webpage.

Problem 6 [12 points]: In this question, we examine two commands available on Kali Linux: `nslookup` and `dig`. Open a terminal on Kali Linux, and type:
`nslookup <your-favorite-server-name>` or `dig <your-favorite-server-name>`. For example: `nslookup ku.edu` or `dig ku.edu`. You can visit the same websites as in Problem 5. Answer the following questions. You can include screenshots to support your answers.

(a) Describe the purpose of these commands. Do you observe any differences between responses returned by these commands.

Solution [3 points]: nslookup (Name Server lookup) and dig (Domain Information Groper) both are command-line tools used for querying DNS to retrieve information about domain names, IP addresses, and other DNS records. However, as shown in Fig.3 and Fig.4, dig is more detailed than nslookup and is commonly used by network administrators for DNS troubleshooting.

```
┌──(kali㉿kali)-[~]
└─$ nslookup www.wireshark.org
Server:         192.168.160.2
Address:        192.168.160.2#53

Non-authoritative answer:
Name:   www.wireshark.org
Address: 172.67.75.39
Name:   www.wireshark.org
Address: 104.26.10.240
Name:   www.wireshark.org
Address: 104.26.11.240
Name:   www.wireshark.org
Address: 2606:4700:20::681a:af0
Name:   www.wireshark.org
Address: 2606:4700:20::681a:bf0
Name:   www.wireshark.org
Address: 2606:4700:20::ac43:4b27
```

Figure 3: Screenshot for nslookup command

```
┌──(kali㉿kali)-[~]
└─$ dig wireshark.org

; <<>> DiG 9.20.4-3-Debian <<>> wireshark.org
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 27775
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;wireshark.org.                 IN      A

;; ANSWER SECTION:
wireshark.org.          5       IN      A       172.67.75.39
wireshark.org.          5       IN      A       104.26.11.240
wireshark.org.          5       IN      A       104.26.10.240

;; Query time: 20 msec
;; SERVER: 192.168.160.2#53(192.168.160.2) (UDP)
;; WHEN: Fri Feb 28 23:05:52 EST 2025
;; MSG SIZE  rcvd: 90
```

Figure 4: Screenshot for dig command

(b) Report a few options and their use cases available for `nslookup`. To explore different options available, you can type `man nslookup` in the terminal.

Solution [3 points]: nslookup provides several options to customize DNS queries. By default, nslookup retrieves A records (IPv4 addresses). To retrieve other records, we can specify the type of record. For example, for IPV6, we can specify option as -type=AAAA. For retrieving mail exchange information , we can specify -type=MX. Also, we can use -timeout=N to change the query time settings to Set timeout (in seconds) for response.

(c) To explore different options available for dig command, type `man dig`. Report a few options and their purpose. To use an option, the syntax is as follows: `dig <server-name> <option>`. For example, specify option "ANY" for two different websites and report your observations.

Solution [3 points]: Similar to nslookup, dig queries type A records (IPv4) by default. However, we can specify different record types, by specifying the option. For example, dig wireshark.org NS to query for name servers or dig wireshark.org AAA to query for IPV6. Also, dig uses UDP by default. However, we can force dig to use TCP using +tcp as an option. For example, dig wireshark.org +tcp.

(d) Explore the command `dig <server-name> +trace`. Explain how trace option works.

Solution [3 points]: As shown in Fig.5 +trace option in dig allows you to trace the entire path of a DNS query, starting from the root DNS servers and working down to the authoritative DNS server that provides the answer. This is useful for troubleshooting DNS resolution problems and understanding how DNS queries propagate through different DNS servers.

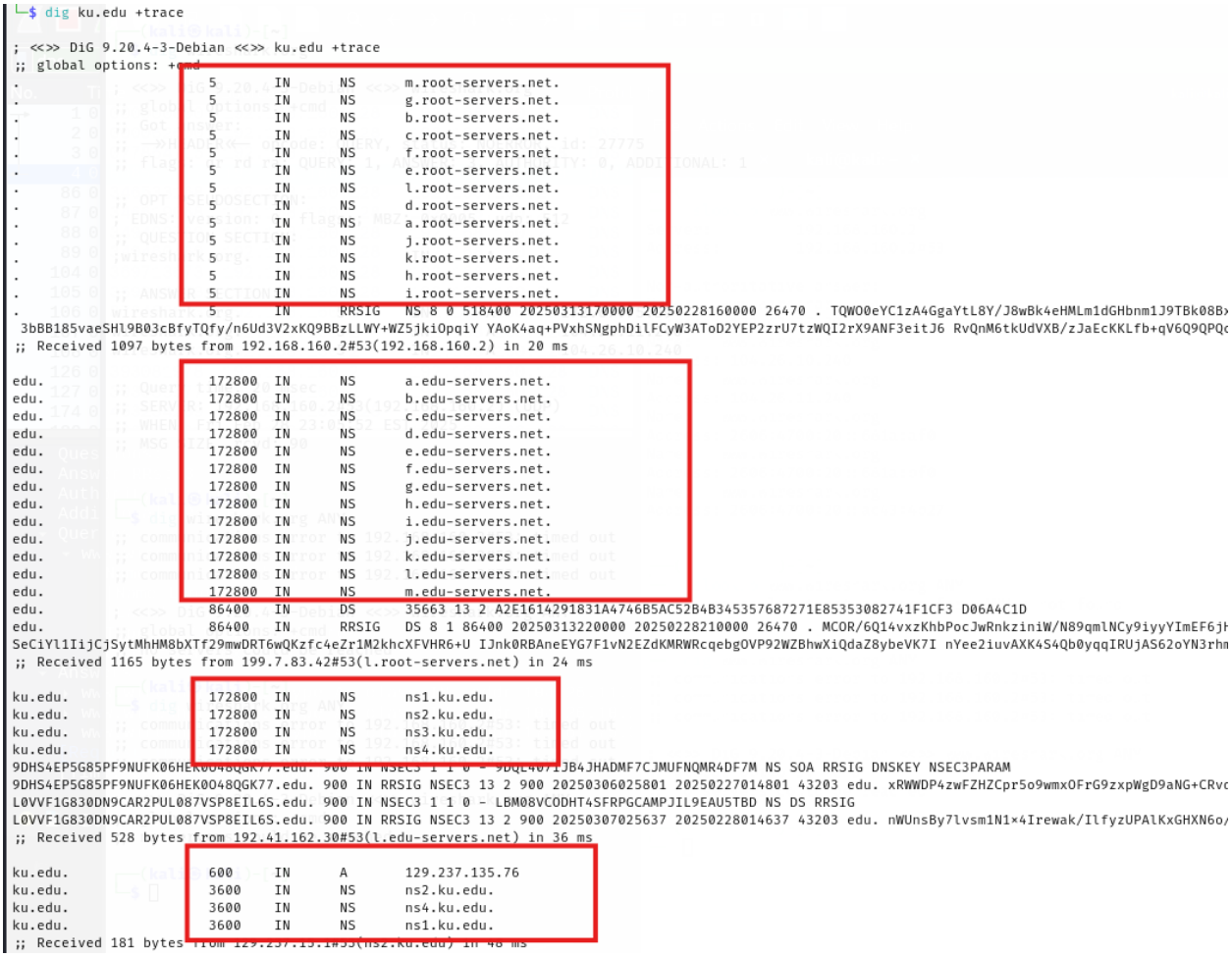


Figure 5: Screenshot for dig command with +trace for ku.edu